# Module 8

## Supporting Management Network

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# Learning Objectives

- Explain the use of configuration and change management documentation.
- Use discovery and monitoring tools to identify network assets.
- Use event management to ensure network availability.
- Use packet analysis and traffic metrics to troubleshoot performance issues.

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# Lesson 8.1

## Organizational Policies and Documentation

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# Policies and Documentation

## Importance of documentation

- Facilitates troubleshooting
- Ensures consistency
- Supports scalability and upgrades
- Supports staff overturn

## Types of documentation

- Configuration management
- Backup management
- Change management
- Asset management
- Network management

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# Configuration Management

- Identify service assets
- Consider a CMS solution
- Determine an identification strategy
- Establish a CI management plan
- Monitor configuration drift

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# Network Device Backup Management

- Document backups and procedures
- Maintain a regular backup schedule
- Audit and verify backups
- Maintain version history
- Configure remote logging of state data

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# Change Management

- Establish a comprehensive documentation protocol
- Ensure consistent use of templates
- Implement version control and access management
- Incorporate a feedback loop
- Regularly review and update documentation

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# Asset Management - Inventory

Update inventory documentation regularly

Record asset description, purchase date, service history, status, and location

Adopt asset management software tools

Implement strict access controls to inventory documentation

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# Network Management

## Physical network diagrams

- Detail hardware components
- Record location information
- Specify cabling details

## Logical network diagrams

- Display protocols being used
- Organize by function vs physical location
- Identify interconnection points

## IP address management

- Use a consistent addressing scheme
- Record IP addresses
- Use automation tools

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# Activity: Worst Case Scenario

What if there wasn’t documentation and….

- The network administrator left the company?
- There was a natural disaster?
- Primary systems crashed?

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# Lesson 8.2

## Host Discovery and Monitoring

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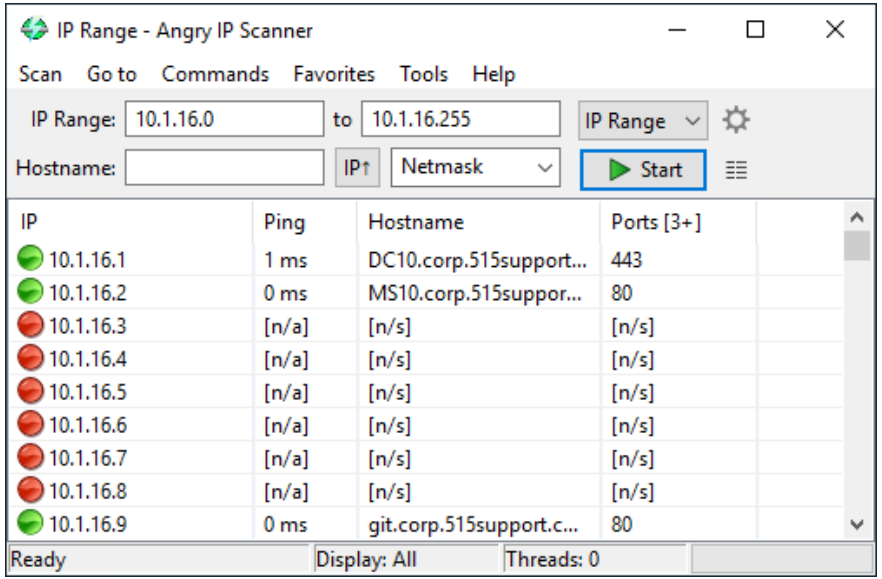
# Network Discovery

## Network Discovery

- Identifying network devices and services
- Network management and security auditing

## Network Discovery Tools

- Nmap
- AngryIP
- PRTG

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# Nmap Scanning Techniques

## Basic Scans

- Default action pings and sends TCP ACK packets to ports 80 and 443
- ARP and Neighbor Discovery (ND) sweeps on local networks

## Host Discovery

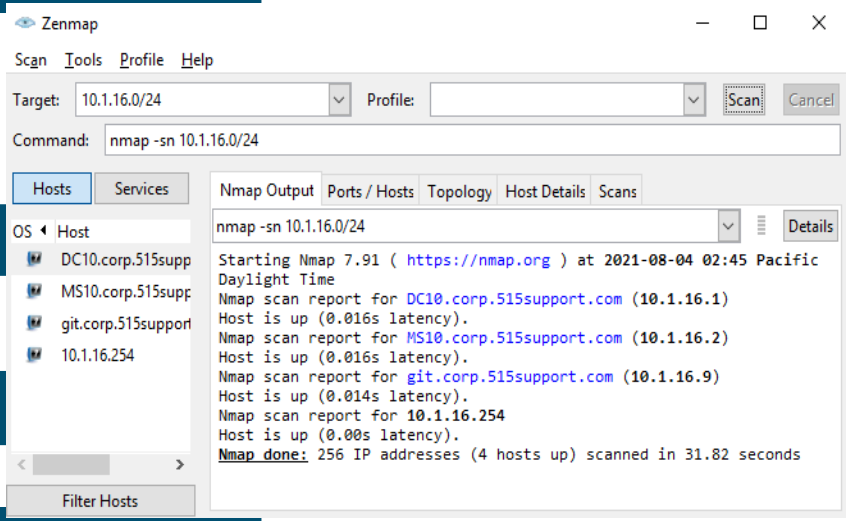
- –sn switch performs discovery without port scanning

## Service and OS Detection

- Identify services and OS running on a host

## Stealth Scans

- Evade detection
- Identify non-responsive hosts

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# Discovery Protocols

## Cisco Discovery Protocol (CDP)

- Cisco proprietary tool
- Discovers how devices are connected in a network
- Discovers OS version and IP addresses
- Detects information from directly connected CDP devices

## Link Layer Discovery Protocol (LLDP)

- IEEE standards-based protocol
- Discovers how devices are connected in a network
- Discovers OS version and IP addresses
- Detects information from all directly connected LLDP devices

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# Performance Monitoring

Tracks and analyzes the speed and efficiency of a network

| Metrics tracking | Baseline establishment | Threshold alerts |
| --- | --- | --- |
| • Bandwidth<br>• Throughput<br>• CPU and Memory<br>• Storage<br>• Latency<br>• Response Time<br>• Error Rate | • Based on historical value<br>• Compared to current performance | • Ensures optimal system performance<br>• Alerts when metrics deviate |

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# Availability Monitoring

Verifies that network devices and services are operational and accessible when needed

- Early detection of outages
- Preventing wider impact
- Optimize server performance
- Network stability
- Security threats
- External validation tools

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# Configuration Monitoring

Verifies that all network appliances are in a known state

- Baseline or golden configuration
- Production configuration
- Backup configuration

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# Activity: Fill in the Blank

1. _____ monitoring verifies that network devices and services are operational and accessible when needed.
2. _____ monitoring verifies that all network appliances are in a known state.
3. _____ monitoring tracks and analyzes the speed and efficiency of a network.

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# Lesson 8.3

## SNMP

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# Simple Network Management Protocol

## Simple Network Management Protocol (SNMP)

- Operates at the application layer
- Uses UDP ports 161/162

## SNMP Functions

- Monitor network devices
- Detect network faults
- Configure remote devices

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# Components of SNMP

## SNMP Manager (NMS)

- Centralized system for network monitoring
- Also known as a Network Management Station

## SNMP Agent

- Software module on managed devices (e.g., routers, switches)
- Maintains information in a database

## Management Information Base (MIB)

- Hierarchical collection of managed resource information
- Organized into categories (e.g., system, interface, IP, and more)

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# SNMP Messages

**GetRequest**

Retrieves data from SNMP agents

**GetNextRequest**

Gets the value of a variable (used for table entries)

**SetRequest**

Sets the value of an object instance

**Response**

Confirms successful value setting

**Trap**

Sent by agents when a fault occurs

**InformRequest**

Like a trap but with acknowledgment

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# SNMP Security

## SNMP v2c Security Tips

- Avoid transmitting plaintext community strings over networks.
- Use complex community strings; avoid defaults.
- Restrict operations via access control lists to known IPs.

## SNMP v3 Advancements

- Offers encryption and strong user authentication
- Uses username lists with access permissions instead of community strings

## Auth modes

- **authNoPriv**: Authentication without encryption
- **authPriv**: Authentication with encryption using user credentials

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# Activity: True or False?

CompTIA

An SNMP Manager is a hierarchical collection of managed resource information that is organized into categories like system, interface, IP, and more.

**True? or False?**

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# Lesson 8.4

## Event Management

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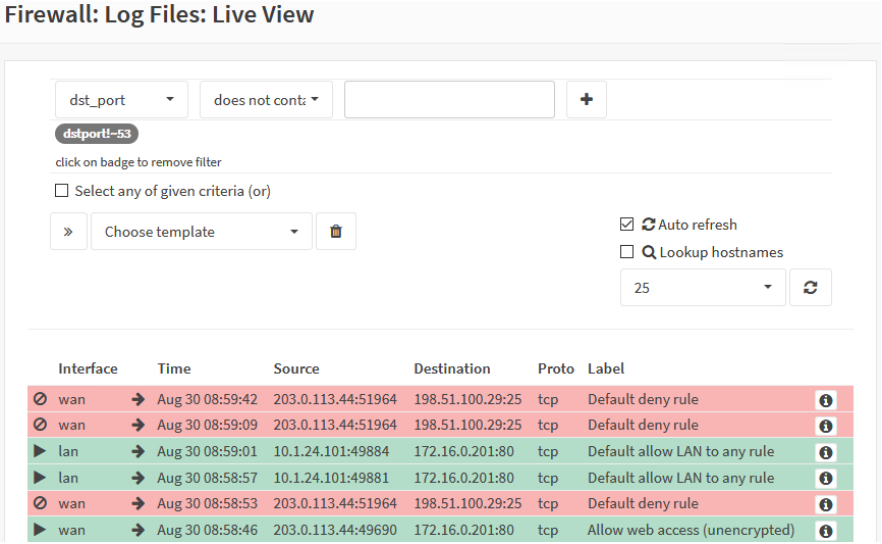
# Network Device Logs

## Network Device Logs Defined

- Data sources for network monitoring, troubleshooting, security audits
- Metadata

## Key Log Types

- System Logs
- Application Logs
- Audit Logs
- Performance/ Traffic Logs

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# Network Device Log Usage

## Troubleshooting & Performance

- Pinpoint network issues
- Optimize performance

## Security & Compliance

- Track unauthorized access, breaches
- Critical for security policies, regulations compliance

## Log Management Practices

- Regular review and analysis for issue prevention
- Secure storage for data integrity and forensic use

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# Log Collectors

## Objective

- Centralize, simplify network log management

## How It Works

- Aggregate log data into single repository

## Benefits

- Centralized Management
- Efficiency
- Scalability

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# Syslogs

## Objective

- Provide a standardized protocol for sending log messages

## Key Features

- UDP Port 514:
- PRI Code
- Flexibility

## Advantages

- Widespread adoption
- Simplifies integration

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# Event Prioritization and Alerting

## Event Prioritization

- Determine event importance from devices
- Categorize events, Emergency (0) to Debug (7)
- Identify immediate action vs. informational events

## Alerting

- Automated alerts based on event severity
- Triggered by event types, thresholds, or anomalies
- Alerts need attention; notifications inform

## Why It Matters

- Prevents performance issues
- Filters out non-critical alerts
- Prioritizes critical events for system integrity

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# Syslog Severity Levels

| Code | Level | Interpretation |
|---|---|---|
| 0 | Emergency | The system is unusable (kernel panic). |
| 1 | Alert | A fault requiring immediate remediation has occurred. |
| 2 | Critical | A fault that will require immediate remediation is likely to develop. |
| 3 | Error | A nonurgent fault has developed. |
| 4 | Warning | A nonurgent fault is likely to develop. |
| 5 | Notice | A state that could potentially lead to an error condition has developed. |
| 6 | Informational | A normal but reportable event has occurred. |
| 7 | Debug | Verbose status conditions used during development and testing. |

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# Event Prioritization and Alerting

## Effective Alert Management Strategies

- Triage process for alerts (Immediate, medium, and low-priority)
- Custom alert thresholds

## Streamlining Alert Responses

- Automation tools for common alerts/ standard responses
- Incident Management Systems integration

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# SIEM Overview

## Definition

- Analyzes security alerts from applications and network devices in real-time

## Purpose

- Integrates security information management (SIM) and security event management (SEM)

## Key Functions

- Log Aggregation
- Event Correlation
- Alerting

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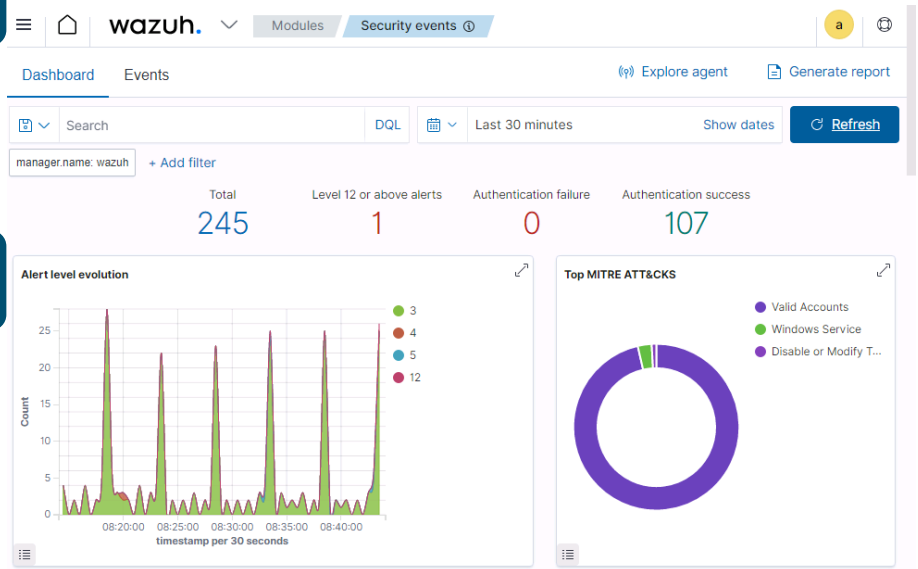
# SIEM Overview

## Event Management Capabilities

- Real-time visibility
- Threat detection and response
- Compliance management

## Implementation Benefits

- Enhanced security posture
- Reduced incident response time
- Improved efficiency

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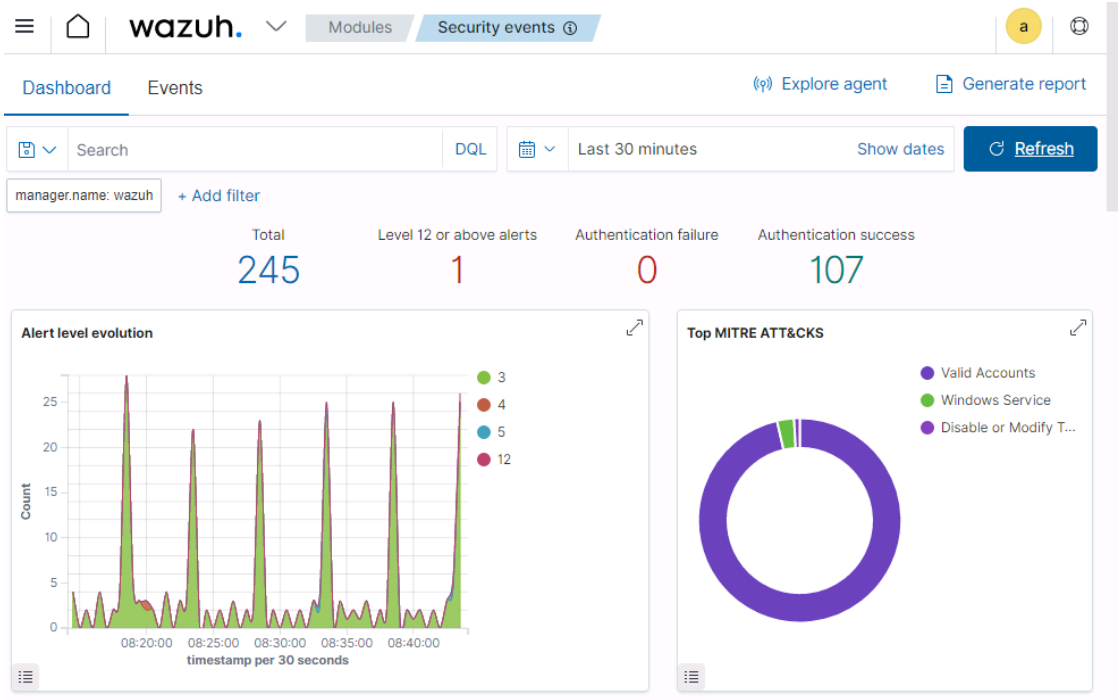
# SIEM Example

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# Activity: Matching

| Audit Logs | Application Logs | System Logs | Performance/ Traffic Logs |
|---|---|---|---|

| Metrics for compute, storage, network | OS events, configuration, kernel processes | Service-specific data (DNS, HTTP) | Authentication and authorization attempts |
|---|---|---|---|

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# Lesson 8.5

## Packet Capture and Analysis

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# Packet Capture

## Definition & Purpose

- Recording network traffic for analysis and troubleshooting

## Key Concepts

- Use libpcap library for capturing packets
- Filtering capabilities to capture specific data

## Practical Application

- Demonstrating how to initiate a packet capture session and important command lines (e.g., `tcpdump -i eth0`)

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# Packet Analysis Tools

## Overview

- Tools that assist in analyzing captured network packets
- Used to diagnose issues or monitor network health.

## Featured Tools

- tcpdump
- Wireshark
- ngrep

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# Packet Capture Analyzer Example

Capturing from Ethernet

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/> Expression...

| No. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 4818 | 363.499645 | 10.1.0.2 | 10.1.0.1 | DNS | 84 | Standard query 0xc36a AAA |
| 4819 | 363.499679 | 10.1.0.2 | 10.1.0.1 | DNS | 84 | Standard query 0x03ee A s |
| 4820 | 363.502559 | 10.1.0.102 | 10.1.0.2 | IMAP | 68 | Request: 1 capability |
| 4821 | 363.509073 | 10.1.0.2 | 10.1.0.102 | IMAP | 162 | Response: * CAPABILITY IM |
| 4822 | 363.515890 | 10.1.0.102 | 10.1.0.2 | IMAP | 95 | Request: 3 login "sam@515 |
| 4823 | 363.520309 | 10.1.0.2 | 10.1.0.1 | TCP | 66 | 49750 → 88 [SYN, ECN, CWR |
| 4824 | 363.520574 | 10.1.0.1 | 10.1.0.2 | TCP | 66 | 88 → 49750 [SYN, ACK, ECN |
| 4825 | 363.520591 | 10.1.0.2 | 10.1.0.1 | TCP | 54 | 49750 → 88 [ACK] Seq=1 Ac |
| 4826 | 363.520608 | 10.1.0.2 | 10.1.0.1 | KRB5 | 277 | AS-REQ |
| 4827 | 363.521861 | 10.1.0.1 | 10.1.0.2 | KRB5 | 244 | KRB Error: KRB5KDC_ERR_PR |
| 4828 | 363.521926 | 10.1.0.2 | 10.1.0.1 | TCP | 54 | 49750 → 88 [FIN, ACK] Seq |

```
> Frame 4822: 95 bytes on wire (760 bits), 95 bytes captured (760 bits) on interface 0
> Ethernet II, Src: Microsof_01:ca:94 (00:15:5d:01:ca:94), Dst: Microsof_01:ca:92 (00:15:5d:01:ca:92)
> Internet Protocol Version 4, Src: 10.1.0.102, Dst: 10.1.0.2
> Transmission Control Protocol, Src Port: 1129, Dst Port: 143, Seq: 15, Ack: 124, Len: 41
v Internet Message Access Protocol
  v Line: 3 login "sam@515support.com" "Pa$$w0rd"\r\n
      Request Tag: 3
      Request Command: login
      Request: login "sam@515support.com" "Pa$$w0rd"

0030  02 00 61 06 00 00 33 20  6c 6f 67 69 6e 20 22 73   ··a···3 login "s
0040  61 6d 40 35 31 35 73 75  70 70 6f 72 74 2e 63 6f   am@515su pport.co
0050  6d 22 20 22 50 61 24 24  77 30 72 64 22 0d 0a      m" "Pa$$ w0rd"··
```

Remainder of request line (imap.request), 37 bytes | Packets: 59212 · Displayed: 59212 (100.0%) | Profile: Default

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# Capture Analysis Techniques

## Analysis Objectives

- Understanding traffic flow
- Identifying misconfigurations
- Detecting anomalies

## Wireshark Analysis Features

- Frame-by-frame header and payload examination
- Use of Follow TCP Stream to reconstruct session data

## Statistical Tools

- Conversations and Protocol Hierarchy
- Traffic analysis

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# Activity: Quick Search

## Take a few minutes to research these tools

- Tcpdump
- Wireshark
- ngrep

## Share your findings with the class

- What is the tool used for?
- How does it help with network management?

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# Lesson 8.6

## Traffic Monitoring

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# Traffic Monitoring

| Definition | Key Points |
| --- | --- |
| • Continuously observing and analyzing the flow of traffic across a network to ensure optimal performance and security | • Identifies traffic volume trends<br>• Monitors performance to detect anomalies<br>• Helps in capacity planning and network design adjustments |

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# Common Performance Issues

Typical problems that affect network efficiency and user experience

## Types of Common Issues

- Packet loss, delays, and jitter affecting quality of service (QoS)
- Bandwidth bottlenecks leading to slow data transfer rates
- Misconfigured network hardware
- Outdated infrastructure

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# Flow Data

Information extracted from data packets that provides insights into the traffic flow within a network

## Key Points

- Includes source/destination IPs, packet sizes, and timestamps
- Essential for network performance analysis and troubleshooting
- Used in traffic profiling and anomaly detection

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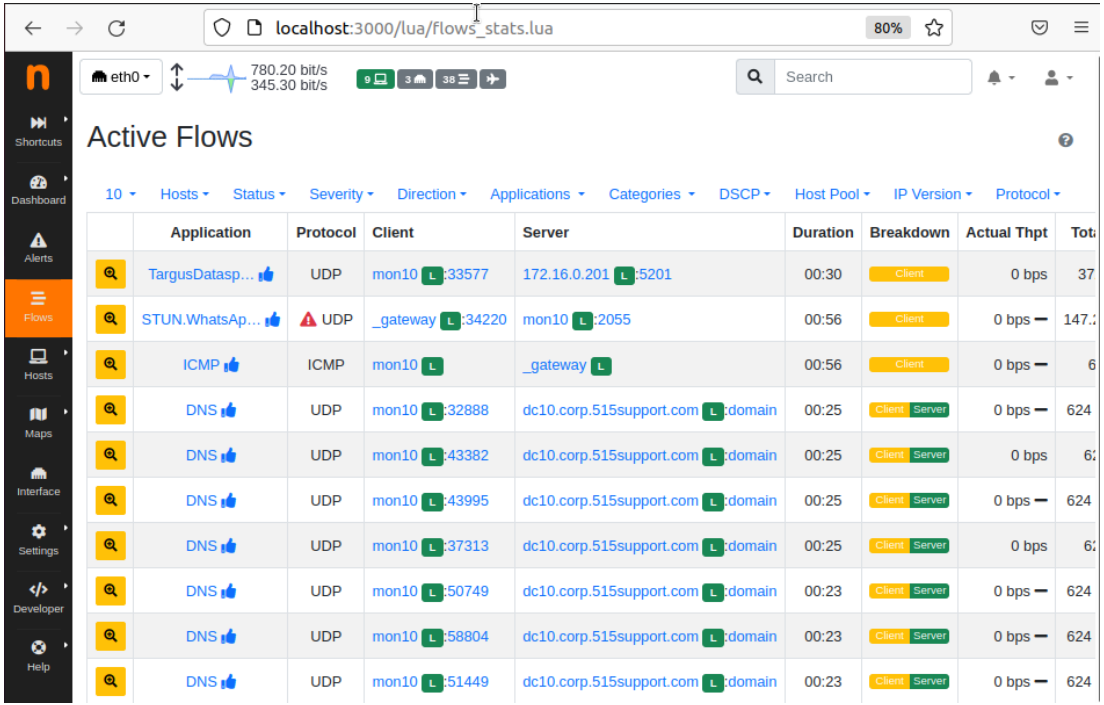
# Monitoring Flow Data Example

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# Traffic Testing Tools

| Definition | Key Points |
| --- | --- |
| • Applications that simulate network traffic and test the performance of network components | • Identify network bottlenecks and capacity limits<br>• Includes packet generators, network emulators, and throughput testers<br>• Examples: Wireshark, iperf, and NetFlow Analyzer |

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# Traffic Testing Tools Examples

```
mon@mon10:~/Desktop$ iperf3 -c 172.16.0.201 -u -b100M -t 30 -i 10
Connecting to host 172.16.0.201, port 5201
[  5] local 10.1.16.242 port 35889 connected to 172.16.0.201 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-10.00  sec   119 MBytes   100 Mbits/sec  86319
[  5]  10.00-20.00  sec   119 MBytes   100 Mbits/sec  86325
[  5]  20.00-30.00  sec   119 MBytes   100 Mbits/sec  86326
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datag
rams
[  5]   0.00-30.00  sec   358 MBytes   100 Mbits/sec  0.000 ms  0/258970 (0%)  s
ender
[  5]   0.00-30.00  sec   358 MBytes   100 Mbits/sec  0.016 ms  0/258970 (0%)  r
eceiver

iperf Done.
```

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# Bandwidth Management

## Definition

- Techniques to control traffic flow in a network to optimize or guarantee performance

## Key Points

- Allocates bandwidth so essential services have priority
- Prevents network congestion and ensures fair usage
- Methods: Rate limiting and traffic policing

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# Traffic Shaping

## Definition

- Prioritizing network traffic to ensure critical applications receive their required bandwidth

## Key Points

- Delaying packets to regulate traffic flow and reduce congestion
- Tools for traffic management: QoS, DiffServ, and MPLS
- Ensures high priority services maintain performance

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# Activity: Two Truths and a Lie

Bandwidth bottlenecks lead to slow data transfer rates.

Bandwidth management is used to control traffic flow in a network to optimize or guarantee performance.

Bandwidth management tools include QoS, DiffServ, and MPLS.

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# Summary

- Implement configuration and change management practices
- Maintain a detailed network asset inventory with diagrams
- Deploy network analyzers for performance and activity insights
- Configure endpoints for log collection
- Define metrics to monitor network health, traffic, and device performance

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